

Technical Report

Hybrid Quantum–Classical Optimization for Call-Center Staffing under Dual Service-Level and Answer-Time Constraints

Warm-Start CVaR-QAOA · Pauli Correlation Encoding · Exact Classical Baselines

Abstract

This report documents a complete hybrid quantum–classical solution to the call-center staffing problem. The objective is to minimize total staffing and overtime cost subject to simultaneous targets on service level ($\text{SLA} \geq 80\%$) and average speed of answer ($\text{ASA} \leq 25\text{ s}$). Decision variables are unary binary encodings of agents per shift; interval coverage is linear in headcounts; quality metrics follow Erlang-C. A slack-free dual QUBO is solved by (i) warm-start Conditional Value-at-Risk QAOA and (ii) Pauli Correlation Encoding, which compresses twelve master variables to five qubits. Classical repair, one-flip search, multi-start local search, linear ILP, and exact enumeration form a baseline ladder. On the standard synthetic instance the exact dual-feasible optimum is \$1496 with plan $n^* = [4, 3, 2]$; both hybrid pipelines recover this cost after post-processing, while greedy is approximately 12% more expensive. We state the formulation, algorithms, verification protocol, results, limitations, and a scalability path aligned with the challenge judging axes of optimality, speed, and scale.

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1. Introduction

Modern contact centers operate under strict contractual quality metrics and rising labor cost pressure. Planners must decide how many agents to assign to each shift so that forecasted interval demand is served at acceptable speed and service level, while overtime and idle time remain controlled. The problem is combinatorial: shift structures, breaks, skill or cover incidence, and nonlinear queueing metrics interact, so constructive heuristics may be dual-feasible yet far from minimum cost.

The challenge scoring rule used in this work is explicit:

Among staffing plans that achieve $SLA \geq 80\%$ and $ASA \leq 25$ seconds under simulation, the plan with the lowest total staffing and overtime cost scores highest.

Judging also considers speed and scalability. Our response is a hybrid pipeline that (1) formulates the dual-constrained integer program, (2) relaxes it to a slack-free QUBO, (3) attacks the QUBO with warm-start CVaR-QAOA and with PCE, (4) restores dual feasibility by classical repair, and (5) benchmarks every method against an exact dual-feasible optimum on tractable instances.

Contributions.

- A dual-metric formulation with unary variables, Erlang requirement tables, and a true non-linear acceptance filter.
- Slack-free shortfall-squared QUBO / Ising embedding without capacity slack qubits.
- Two quantum hybrids: CVaR-QAOA with classical warm start, and PCE reducing 12 variables to 5 qubits.
- A classical ladder including exact enumeration as ground truth under the scoring rule.
- Multi-seed verification and an honest gap analysis relative to exact, not only relative to greedy.

2. Problem setting and notation

2.1. Operational structure

Let $\mathcal{S} = \{1, \dots, S\}$ be the set of shifts and $\mathcal{T} = \{1, \dots, T\}$ the set of planning intervals (e.g., thirty-minute buckets). Incidence $C_{s,t} \in \{0, 1\}$ indicates whether shift s covers interval t ; breaks are encoded by zeros on the break interval. Each shift has a unit cost $c_s > 0$; overtime shifts are scaled by a multiplier ρ_{OT} (typically 1.5).

2.2. Demand and offered load

Forecasted arrivals in interval t are denoted $\lambda_t \geq 0$. With interval length Δ (seconds) and average handle time H (seconds), the offered load in Erlangs is

$$A_t = \frac{\lambda_t}{\Delta} H. \quad (1)$$

2.3. Decision variables

Unary binary variables $x_{s,k} \in \{0, 1\}$ for $k \in \{0, \dots, N_{\max} - 1\}$ encode headcount

$$n_s = \sum_{k=0}^{N_{\max}-1} x_{s,k} \in \{0, \dots, N_{\max}\}. \quad (2)$$

Coverage is $\text{cov}_t = \sum_s C_{s,t} n_s$. The binary width is $N = S \cdot N_{\max}$.

Remark 1. *Unary encoding keeps penalties linear in headcount before coverage expansion and simplifies warm-start from integer classical solutions. Log encodings are possible but add bits and decoding complexity.*

3. Erlang-C quality model

3.1. Waiting probability

For n agents and load A , the Erlang-C probability that a call waits is

$$C(n, A) = \begin{cases} \frac{A^n n / (n!(n-A))}{\sum_{k=0}^{n-1} A^k / k! + A^n n / (n!(n-A))} & n > A, \\ 1 & n \leq A. \end{cases} \quad (3)$$

3.2. Service level and ASA

$$\text{SL}(n, \lambda) = 1 - C(n, A) \exp(-(n-A)\tau/H), \quad (4)$$

$$\text{ASA}(n, \lambda) = \frac{C(n, A) H}{n - A} \quad (n > A), \quad (5)$$

with τ the contractual answer-time threshold. Occupancy is $\min(1, A/n)$. Abandonment is modeled as an increasing function of ASA.

3.3. Requirement inversion

$$R_t^{\text{SLA}} = \min\{n \in \mathbb{Z}_+ : \text{SL}(n, \lambda_t) \geq \text{SL}^*\}, \quad (6)$$

$$R_t^{\text{ASA}} = \min\{n \in \mathbb{Z}_+ : \text{ASA}(n, \lambda_t) \leq \text{ASA}^*\}. \quad (7)$$

Both are monotone in the relevant regime and computed by sequential search from $\lceil A_t \rceil$.

3.4. True dual filter

After a candidate $\mathbf{n}$ is obtained, call-weighted averages

$$\text{SLA}(\mathbf{n}) = \frac{\sum_t \lambda_t \text{SL}(\text{cov}_t, \lambda_t)}{\sum_t \lambda_t}, \quad (8)$$

$$\text{ASA}(\mathbf{n}) = \frac{\sum_t \lambda_t \text{ASA}(\text{cov}_t, \lambda_t)}{\sum_t \lambda_t} \quad (9)$$

define dual feasibility: $\text{SLA} \geq \text{SL}^*$ and $\text{ASA} \leq \text{ASA}^*$. Linear covers using R are sufficient but not necessary for this filter.

4. Integer program and classical baselines

4.1. Primal IP

$$\min_{\mathbf{n}} \quad \sum_s c_s n_s \quad (10)$$

$$\text{s.t.} \quad \text{cov}_t \geq R_t^{\text{SLA}}, \quad \text{cov}_t \geq R_t^{\text{ASA}}, \quad 0 \leq n_s \leq N_{\max}. \quad (11)$$

4.2. Exact enumeration

The box $\{0, \dots, N_{\max}\}^S$ has size $(N_{\max} + 1)^S$. For $S = 3$, $N_{\max} = 4$ this is 125 vectors. Evaluating dual feasibility and cost on every vector yields the exact scored optimum $\mathbf{n}^*$. This is the primary classical ground truth in our experiments.

4.3. Linear ILP surrogate

Unary binaries with cover $\text{cov}_t \geq \max(R_t^{\text{SLA}}, R_t^{\text{ASA}})$ are solved by branch-and-cut (CBC) and re-scored with Erlang. Because hard R ignores curvature of SL/ASA, the ILP optimum can be strictly more expensive than $\mathbf{n}^*$.

4.4. Greedy constructive heuristic

Residual requirement $\mathbf{r} = \max(\mathbf{R}^{\text{SLA}}, \mathbf{R}^{\text{ASA}})$ is reduced by repeatedly selecting the shift that covers the most positive residual per unit cost, then applying SLA/ASA repair. Greedy is fast and often dual-feasible but suboptimal.

4.5. Multi-start local search

Starts from greedy and random integer vectors; repair to dual feasibility; drop agents when the reduced plan remains dual-feasible. On the standard instance this recovers $\mathbf{n}^*$ across all tested

seeds.

5. Slack-free QUBO and Ising form

Hard covers are softened by shortfall-squared penalties:

$$h(x) = \sum_s \sum_k c_s x_{s,k} + P_{\text{SLA}} \sum_t \lambda_t^{\text{dual}} (\text{cov}_t - R_t^{\text{SLA}})^2 + P_{\text{ASA}} \sum_t w_t (\text{cov}_t - R_t^{\text{ASA}})^2. \quad (12)$$

Expanding $(\text{cov} - R)^2$ and using $x_i^2 = x_i$ produces a QUBO polynomial without auxiliary slack variables. The Ising Hamiltonian

$$H = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j + E_0 \quad (13)$$

follows from $x_i = (1 - Z_i)/2$. Optional dual multipliers λ_t^{dual} are updated by dual averaging on residual shortfalls between outer iterations.

6. Warm-start CVaR-QAOA

6.1. Ansatz and warm start

A p -layer QAOA state is prepared from a product state whose R_Y angles are derived from a softened classical bitstring $\tilde{x}_i \in (\varepsilon, 1 - \varepsilon)$ (typically greedy). Cost layers implement $e^{-i\gamma H}$ and mixers $e^{-i\beta B}$.

6.2. CVaR training objective

For risk level $\alpha \in (0, 1]$,

$$\text{CVaR}_\alpha(E) = \frac{1}{\alpha} \int_0^\alpha F_E^{-1}(u) du \quad (14)$$

is minimized over (γ, β) by a derivative-free optimizer (COBYLA). This focuses the variational loop on the low-energy tail of the measurement distribution rather than on mean energy alone.

6.3. Sampling and acceptance

M bitstrings are drawn, each repaired for coverage and ASA, and dual-filtered. The retained plan is the dual-feasible sample of minimum cost (or a declared failure if none exist).

7. Pauli Correlation Encoding

7.1. Capacity and compression

Degree- D Z -correlators on n_q qubits number $\sum_{d=1}^D \binom{n_q}{d}$. The minimal width supporting N variables is

$$n_q^*(N, D) = \min \left\{ q : \sum_{d=1}^D \binom{q}{d} \geq N \right\}. \quad (15)$$

For $N = 12$ and $D = 2$, $n_q^* = 5$ because $\binom{5}{1} + \binom{5}{2} = 15 \geq 12$, a reduction of about 58%.

7.2. Training and decode

An EfficientSU2 ansatz on n_q qubits is trained so soft bits

$$x_i^{\text{soft}} = \frac{1}{2} (1 - \tanh(\alpha \langle P_i \rangle)) \quad (16)$$

minimize $h(x^{\text{soft}})$. Hard bits use $\text{sign}(\langle P_i \rangle)$, followed by repair and one-flip local search.

Remark 2. *Raw PCE often lands near greedy; the one-flip post-process is material for matching the exact dual cost. This mirrors classical bit-flip improvement used in related divert / DOM PCE pipelines.*

8. Classical repair operators

Coverage repair. While some interval has $\text{cov}_t < R_t^{\text{SLA}}$, increment an agent on a feasible covering shift with residual capacity.

ASA push. While $\max_t \text{ASA}(\text{cov}_t, \lambda_t) > \text{ASA}^*$, increment similarly on the worst interval.

One-flip search. From a dual-feasible bitstring, accept bit flips that preserve dual feasibility and strictly reduce cost, until a local optimum.

These operators are shared by quantum and random-sampling control arms so that comparisons isolate distribution quality versus post-processing power.

9. Synthetic demand and manager scenarios

Arrivals are Poisson with a bimodal time-of-day profile, day-of-week scales, queue factors, and channel factors (voice heavier than chat). The optimizer uses multi-day interval means. Manager scenarios reweight the same model: reduced penalties (cost mode), increased penalties (service mode), tilted shift costs (preference), and buffered R_t (resilience). Dual feasibility is always enforced at acceptance.

10. Experimental protocol

Standard instance. $T = 12$, $S = 3$, $N_{\max} = 4$ ($N = 12$ bits). Targets $SL^* = 0.80$, $ASA^* = 25$ s. Base cost \$160 with mild shift premiums. Primary seed 7; robustness seeds $\{7, 11, 21, 42, 99\}$.

Quantum configuration. CVaR-QAOA: $p = 1$, $\alpha = 0.25$, warm start, COBYLA, Aer matrix-product-state simulator. PCE: degree 2, EfficientSU2 with two repetitions, multiple random restarts, COBYLA on soft energy.

Metrics. Cost, SLA, ASA, dual flag, gap to exact, agents per shift, runtime, qubit count. Multi-seed tables record dual feasibility and cost wins.

11. Results

11.1. Scorecard on the standard instance

Method	Cost (\$)	Gap to exact	Dual-feasible	Qubits
Exact enumeration	1496	0	Yes	—
Multi-start local search	1496	0	Yes	—
PCE + 1-flip	1496	0	Yes	5
CVaR-QAOA hybrid	1496	0	Yes	12
Greedy	1672	+176	Yes	—
Linear ILP (CBC)	2016	+520	Yes	—

The exact plan is $n^* = [4, 3, 2]$ with $SLA \approx 93.7\%$ and $ASA \approx 12.1$ s, both inside targets.

11.2. Multi-seed behavior

Across five seeds, multi-start LS and post-processed hybrids remain dual-feasible and track the exact dual cost; greedy stays dual-feasible but consistently more expensive. Linear ILP remains conservative.

11.3. Discussion

The correct optimality statement is *match to exact dual-feasible minimum cost*, not merely improvement over greedy. Multi-start LS shows that a strong classical heuristic also reaches exact on this size. PCE’s distinguishing result is equal quality at five qubits. Claims of quantum advantage at 5–12 qubits are not supported: best-of-N random sampling with the same repair can be competitive at large budgets. The value of the hybrid design is a verified dual-feasible optimum, a qubit-efficient encoding option, and a transparent baseline ladder for judges.

12. Limitations

- Instance scale is chosen so exact enumeration is possible; full enterprise problems (many skills, 48 intervals, large $N_{\max}$) require decomposition.
- Synthetic demand is structured but not calibrated to a confidential production trace.
- Post-processing (repair, 1-flip, best-of-N) contributes substantially to final cost; raw quantum decode is not sufficient.
- Simulator results (MPS / statevector-style estimators) do not include device noise unless a separate noise study is run.
- Linear ILP should not be reported as the exact dual optimum.

13. Scalability and deployment path

1. **PCE** keeps qubit growth sublinear in N for fixed correlator degree.
2. **Time or skill blocks** yield small QUBOs coordinated by classical Benders or column-generation style outer loops.
3. **Slack-free penalties** avoid logarithmic slack registers for capacity.
4. **Warm-start** recycles operational plans as quantum initial states for daily re-optimization.
5. **Manager modes** expose cost–service trade-offs without changing the dual gate.

14. Judging alignment

Optimality Dual-feasible plans at exact minimum cost \$1496 on the verified instance; gaps reported versus exact.

Speed Exact/LS on the small box run in milliseconds; hybrid quantum steps run in about 0.3–2 s under MPS simulation in our tests.

Scalability PCE 12→5 qubits; slack-free encoding; documented decomposition path for larger T and S .

15. Deliverables

Item	Description
Classical baselines notebook	Exact, ILP, LS, greedy, multi-seed plots
PCE notebook	Mapping, training, post-process, scaling
Hybrid CVaR notebook	Dual QUBO, QAOA, manager scenarios
Mathematical formulation PDF	Full equation set
Beamer deck	Ten-frame presentation summary
Source bundle	Python solvers, notebooks, figures, JSON

16. Detailed QUBO coefficient derivation

Let $\mathcal{I}_t$ be the index set of unary bits covering interval t . Then $\text{cov}_t = \sum_{i \in \mathcal{I}_t} x_i$ and

$$(\text{cov}_t - R_t)^2 = \sum_{i \in \mathcal{I}_t} x_i^2 + 2 \sum_{\substack{i < j \\ i, j \in \mathcal{I}_t}} x_i x_j - 2R_t \sum_{i \in \mathcal{I}_t} x_i + R_t^2, \quad (17)$$

where the identity $x_i^2 = x_i$ was used. Multiplying by penalty weight P and summing over t contributes

$$Q_{ii} \leftarrow Q_{ii} + P(1 - 2R_t) \quad \text{for each } i \in \mathcal{I}_t, \quad (18)$$

$$Q_{ij} \leftarrow Q_{ij} + 2P \quad \text{for each unordered pair } \{i, j\} \subset \mathcal{I}_t. \quad (19)$$

Linear staffing costs $\sum_s c_s \sum_k x_{s,k}$ add c_s to each diagonal entry corresponding to shift s . The resulting symmetric matrix Q defines the QUBO minimized by quantum and classical QUBO solvers.

17. Warm-start angle construction

Given a classical bit $b_i \in \{0, 1\}$ and a softening parameter $\varepsilon \in (0, 1/2)$, set

$$\tilde{x}_i = \text{clip}(b_i, \varepsilon, 1 - \varepsilon), \quad \theta_i = 2 \arcsin \sqrt{\tilde{x}_i}. \quad (20)$$

The product state $\bigotimes_i R_Y(\theta_i)|0\rangle$ has single-qubit marginals equal to $\tilde{x}_i$, biasing QAOA toward the classical neighborhood without collapsing the superposition entirely.

18. CVaR estimation from finite samples

Order measured energies $E_{(1)} \leq \dots \leq E_{(M)}$ with empirical probabilities $p_{(j)}$. Let k be the smallest index with cumulative mass at least α . Then

$$\widehat{\text{CVaR}}_\alpha = \frac{1}{\alpha} \left(\sum_{j=1}^{k-1} p_{(j)} E_{(j)} + \left(\alpha - \sum_{j=1}^{k-1} p_{(j)} \right) E_{(k)} \right). \quad (21)$$

This estimator is used inside each COBYLA evaluation during variational training.

19. PCE correlator inventory for $n_q = 5$

Weight-1 operators: $Z_0, \dots, Z_4$ (five correlators). Weight-2 operators: all $\binom{5}{2} = 10$ pairs $Z_i Z_j$. Total capacity 15, of which 12 are assigned to master variables. The unused three correlators may be reserved for future constraints or left idle.

20. Complexity accounting

Method	Dominant cost	Notes
Exact enum	$O((N_{\max} + 1)^S \cdot T)$ Erlang evals	Exact on small S
Greedy	$O(S \cdot T \cdot N_{\max})$	Single pass + repair
Multi-start LS	$O(K \cdot S \cdot T \cdot N_{\max})$	K starts
Linear ILP	Branch-and-cut on N binaries	MIP solver dependent
CVaR-QAOA	$O(I \cdot M \cdot \text{cost}(H))$	I iters, M shots
PCE	$O(I \cdot n_q \cdot \text{cost}(\text{ansatz}))$	Correlator batch estimates

21. Error modes and mitigations

- **Under-coverage after decode:** mitigated by coverage repair and ASA push before scoring.
- **Over-staffing from linear R :** inherent to ILP surrogate; use exact or Erlang-aware search for ground truth.
- **Local optima in COBYLA:** multiple restarts and warm starts reduce variance.
- **Shot noise in CVaR:** increase shots for final evaluation; keep training shots moderate.
- **Decode ambiguity near zero correlators:** increase α sharpness or fall back to soft thresholding with repair.

22. Reproducibility checklist

1. Fix global seeds for NumPy and any quantum sampler backends.

2. Record $P_{\text{SLA}}, P_{\text{ASA}}, \alpha, p, N_{\text{max}}$, shift costs, and Erlang constants (H, Δ, τ) .
3. Persist bitstrings, costs, SLA, ASA, and dual flags per seed.
4. Publish exact enum cost as the reference line in all gap plots.
5. Separate training shots from final evaluation shots in runtime reports.

23. Extended future work

Beyond block decomposition and real traces, several extensions are natural: (i) multi-skill agents with coverage tensors $C_{s,t,g}$ over skill groups g ; (ii) stochastic programming over demand scenarios inside the QUBO expectation; (iii) noise-aware training using fake backends or device calibrations; (iv) integration with workforce management systems via API export of n_s and interval understaffing flags; (v) comparison against commercial constraint solvers on larger unary widths where exact enum is impossible but MIP remains feasible.

24. Worked numerical example

On the standard seed-7 demand vector

$$\lambda = (5, 10, 15, 17, 21, 21, 15, 16, 19, 16, 11, 6),$$

Erlang inversion yields peak $R^{\text{SLA}} = R^{\text{ASA}} = 6$. Exact search returns $n^* = (4, 3, 2)$ with cost

$$4 \cdot 160 + 3 \cdot 168 + 2 \cdot 176 = 640 + 504 + 352 = 1496.$$

Simulated SLA is approximately 93.7% and ASA approximately 12.1 s. Greedy returns $(4, 3, 3)$ at \$1672. PCE with one-flip and CVaR-QAOA with repair both return plans at \$1496 that pass the dual gate. Linear ILP returns $(4, 4, 4)$ at \$2016, illustrating over-coverage relative to nonlinear scoring.

25. Connection to distributed order management (DOM)

The same hybrid pattern—QUBO relaxation, PCE or QAOA sampling, classical feasibility repair—appears in divert recommendation for distribution centers. Staffing replaces divert bits with unary agent slots and replaces capacity docks with Erlang dual filters. Lessons transfer: (i) exact or strong classical references prevent overclaiming, (ii) post-processing is part of the method, not an afterthought, and (iii) qubit compression via correlators is most valuable when master variable counts grow.

26. Summary tables for submission

Primary claim. Hybrid PCE (5 qubits) and CVaR-QAOA (12 qubits) recover the exact dual-feasible minimum staffing cost of \$1496 on the standard instance after classical repair.

Baselines. Exact enumeration defines ground truth; multi-start LS matches it; greedy is +\$176; linear ILP is +\$520.

Risk of overclaiming. Do not claim quantum supremacy at this size. Claim dual-feasible optimality relative to exact, qubit compression via PCE, and a reproducible hybrid pipeline aligned with the scoring rule.

27. Conclusion

We formulated dual-constrained call-center staffing as an integer program with Erlang acceptance, relaxed it to a slack-free QUBO, and solved it with warm-start CVaR-QAOA and Pauli Correlation Encoding. Exact enumeration establishes the dual-feasible minimum cost at \$1496 on the standard instance; both hybrid pipelines recover this optimum after classical repair, matching multi-start local search and improving on greedy by \$176. The technical stand-outs are alignment with the dual scoring rule, an honest exact baseline, and PCE’s reduction from twelve variables to five qubits at equal quality. Future work should validate on real arrival traces and embed the same core in a block-decomposition outer loop for enterprise scale.

A. Symbol summary

Symbol	Meaning
$x_{s,k}, n_s$	Unary bit and integer agents on shift s
$\text{cov}_t, C_{s,t}$	Coverage and shift–interval incidence
λ_t, A_t	Arrivals and offered load
$R_t^{\text{SLA}}, R_t^{\text{ASA}}$	Erlang requirement tables
h, H	QUBO objective and Ising Hamiltonian
CVaR_α	Conditional value-at-risk of energy
P_i, n_q^*	PCE correlator and compressed width
$\mathbf{n}^*, \text{gap}$	Exact dual optimum and cost gap

B. Algorithmic pseudocode sketches

B.1. Exact dual enumeration

1. For each $\mathbf{n} \in \{0, \dots, N_{\max}\}^S$: compute cov, SLA, ASA, cost.
2. If dual-feasible and cost improves incumbent, store $\mathbf{n}$.

3. Return incumbent as $\mathbf{n}^*$.

B.2. CVaR-QAOA hybrid

1. Build dual QUBO / Ising; warm-start angles from greedy bits.
2. Optimize (γ, β) under CVaR_α ; sample M bitstrings.
3. Repair each sample; keep best dual-feasible by cost.

B.3. PCE hybrid

1. Map variables to degree-2 correlators on n_q^* qubits.
2. Train EfficientSU2 on soft energy $h(x^{\text{soft}})$.
3. Hard-decode, repair, one-flip; accept dual-feasible min cost.

Acknowledgments

This technical report accompanies the executable notebooks and source bundle for the call-center staffing challenge track. Numerical values refer to the synthetic standard instance used for exact verification; production deployments should recalibrate Erlang parameters and demand models to site-specific data.